Homework 2

1. Use the bisection method and find the roots of the equation $x^3 - x^2 - x + 1 = 0$ in the respective intervals [-2, 0] and [0,2]. Choose the tolerance ε as 10^{-5}. In each case, before starting the computations, estimate the number of iterations that you might need to find the respective root. Do your actual computations agree with your estimations? Explain your observations.

2. Consider the problem $x = (2 - e^x + x^2)/3$. In which interval it has a solution? Determine this solution using the Newton's method with a tolerance ε as 10^{-5}.

3. Find $(5)^{1/2}$ using the Newton's method.

Due on 1/23/24 in class